

5(a)(i)	$R = \rho L / A$	C1
	$A = (2.6 \times 10^{-8} \times 59) / 3.4 = 4.5 \times 10^{-7} \text{ m}^2$	A1
5(a)(ii)	$I = 1.8 / 3.4$ $= 0.53 \text{ A}$	A1
5(a)(iii)	$I = Anvq$ $v = 0.53 / (4.5 \times 10^{-7} \times 6.1 \times 10^{28} \times 1.60 \times 10^{-19})$	C1
	$= 1.2 \times 10^{-4} \text{ m s}^{-1}$	A1
5(b)(i)	(cross-sectional) area/ A is less	M1
	(I , n , e the same so) average drift speed is greater	A1
5(b)(ii)	(area is less so) more resistance/ R	M1
	(I is the same, so) more power/ P	A1
	or	
	($P = I^2 \rho L / A$ so) $P \propto 1 / A$	(M1)
	(A is less so) more P	(A1)
5(c)(i)	180 Ω and 90 Ω resistors shown connected in parallel	B1
5(c)(ii)	resistors connected in parallel labelled as 180 Ω and 90 Ω and the other resistor labelled as 30 Ω	M1
	V_{OUT} or 8.0 V labelled across the two resistors in parallel	A1